



CSE299 · PROJECT PROPOSAL · NORTH SOUTH UNIVERSITY

CareerSetu

An AI-Powered Alumni – Industry Bridge Platform

Verified profiles · AI CV grooming · smart job matching · scam-free hiring — built for the North South University alumni community.

COURSE & GROUP DETAILS

CSE299 — Project & Technical Writing | Section: 10 | Group: 1

● **Member 1: Jihanur Rahman Ratul - 2412800642**

UI/UX & Frontend (Web + Mobile) Lead

● **Member 3:**

AI / Machine Learning Lead

● **Member 2: Tahsin Ahmed Sufol**

Backend & Database Lead

● **Member 4:**

DevOps, QA & Blockchain Lead

Replace bracketed placeholders with your real section, group number, names and IDs before submission.

What is CareerSetu?

- A unified web & mobile platform that connects North South University alumni, soon-to-graduate students, and employers in one trusted space.
- Alumni build a rich, customizable profile; browse and benchmark other graduates' CVs; and receive AI-driven "CV grooming" feedback to strengthen their own.
- Companies get a dedicated portal to post verified openings — and once an alumnus accepts, instantly access their CV and grade sheet.
- A "Career Magazine" sub-site lets companies run sponsored features and ads, funding the platform alongside paid job postings.
- Three AI/ML systems — CV scoring, job–candidate matching, and job-scam detection — form the platform's "Trust & Growth" engine.

THE BRIDGE WE'RE BUILDING



Alumni & Students

Verified profile + AI-groomed CV



CareerSetu Engine

Matching · Trust Score · Verification



Employers

Ranked, credential-checked talent

A single sign-on ecosystem where trust is engineered in, not assumed.

Why Bangladesh's Job Market Needs This



No structured CV feedback

Students and alumni polish resumes by guesswork with no data-driven feedback loop — a real cost when 29% of unemployed 15–29 year-olds nationally are university graduates (BBS, 2024).



Job scams are rampant

Fake job offers are officially listed among Bangladesh's most common online scams, costing job-seekers money, data, and trust in digital hiring.



Credentials are hard to verify

Employers have no fast way to confirm a candidate's degree, CGPA or certificates — slowing hiring and leaving room for resume fraud.



Alumni networks are fragmented

Alumni, current students and recruiters have no single trusted space to connect, mentor, and hire within the NSU community.

PROBLEM STATEMENT

How might we build a trusted, AI-driven platform where NSU alumni can grow their careers, get discovered by verified employers, and avoid the fraud that plagues Bangladesh's online job market?

Why Now, and What Changes

MOTIVATION

Bangladesh produces hundreds of thousands of graduates every year, yet the market absorbing them is not keeping pace — and job seekers increasingly turn to unverified online channels, where scams thrive.

11–12%

Graduate unemployment rate — nearly 3x the ~4% national average (BBS / The Business Standard, 2025)

29%

Of unemployed youth aged 15–29 nationally hold a university degree (BBS Labour Force Survey, 2024)

Top 5

Fake job offers rank among the most common scam types tracked by Bangladesh's official cyber-scam registry (scamcheck.gov.bd)

EXPECTED OUTCOMES

- Faster, higher-quality job placements through AI-matched openings tailored to each alumnus' profile
- Fewer alumni victimized by job scams, thanks to automatic Trust Scores on every listing
- Verifiable, tamper-proof credentials that speed up employer hiring decisions
- A self-sustaining revenue model (job-posting fees + magazine ads) that funds long-term growth
- Rich, structured alumni-engagement data for NSU's career services office

Where CareerSetu Differs

Capability	Generic Job Boards (e.g. LinkedIn)	Typical University Alumni Portal	CareerSetu
AI CV grooming & match scoring	X	X	✓
Job-post trust / scam detection	Limited	X	✓
Blockchain credential verification	X	X	✓ (optional)
University-specific alumni network	X	✓	✓
Local payment integration (bKash/Nagad)	X	X	✓
Sponsored "career magazine" monetization	X	X	✓

INFORMED BY PRIOR RESEARCH

- Online recruitment fraud detection: Vidros, Kolias, Kambourakis & Akoglu (2017), Future Internet — origin of the EMSCAD fake-job-posting dataset used in Section 13.
- Content-based resume–job matching using NLP embeddings and cosine similarity — an established approach in recent (2021–2025) ML/HR-tech literature.

What We're Building Toward

AIM

To build a trusted, AI-powered career ecosystem that helps NSU alumni grow professionally, while giving employers a fraud-free way to discover verified talent.



Personalized Growth

AI CV grooming that gives every alumnus actionable, individualized feedback.



Verified Trust

Blockchain-backed credentials and ML fraud detection remove guesswork from hiring.



Meaningful Connections

A dedicated space where alumni, students and companies interact directly.



Sustainable Impact

A monetization model that keeps the platform free-to-join and self-funding.

The Alumni Experience



Customizable Profile

Education, skills, certifications, achievements and a shareable public page.



Browse & Benchmark CVs

See how peers in the same field present their experience and skills.



AI CV Grooming & Match Score

NLP-driven feedback and a benchmark score against target roles.



Browse & Apply to Jobs

A searchable, filterable board of verified openings from partner companies.



Personalized Job Recommendations

AI ranks openings by fit to the alumnus' profile and goals.



Post Jobs as an Alumnus

Alumni can list openings at their own company for a small fee — monetization stream 1.

The Company & Admin Experience



Company Dashboard

Branded profile plus a workspace for managing all job postings in one place.



AI-Ranked Candidate Shortlist

Applicants are ranked automatically by fit to the job requirements.



CV & Grade Sheet Access

Full CV and academic record unlocked only after the alumnus accepts.



Trust Score Badge

Every listing is scored by the fraud-detection model before it goes live.



Career Magazine Ads

Sponsored features and banner placements — monetization stream 2.



Admin Moderation Console

Credential verification, listing review, and platform analytics.

How CareerSetu Sustains Itself

REVENUE STREAM 1



Pay-Per-Post & Subscriptions

Companies and alumni pay a listing fee — or subscribe monthly — to publish job openings. Premium tiers unlock AI-ranked shortlists and featured placement at the top of search results.

REVENUE STREAM 2



Career Magazine Ads

A sponsored content and banner-ad sub-site where companies run employer-branding features, success stories and campaigns — billed per placement or campaign.



Future Stream: Verified-Credential API

Once the blockchain trust layer matures, third-party recruiters and background-check agencies can pay per-lookup to verify an alumnus' credentials instantly.

Our Development Approach

- Agile Scrum with 2-week sprints, sprint planning, daily stand-ups and end-of-sprint demos.
- Requirements gathered through a primary survey circulated to NSU alumni, students and partner companies (Google Form), mirroring how similar capstone projects sourced real data.
- Design → Develop → Test → Deploy loop repeated every sprint, with usability feedback folded into the next cycle.
- ML models are trained and evaluated offline (Jupyter/Colab), version-controlled, then served through the AI microservice — retrained periodically as new alumni and job data arrive.
- Every commit triggers an automated CI/CD pipeline: lint → unit test → build → staging deploy.

SPRINT CYCLE



Research

Surveys & literature review



Design

Wireframes & data models



Develop

Feature branches, code review



Test

Unit, integration & E2E tests



Deploy

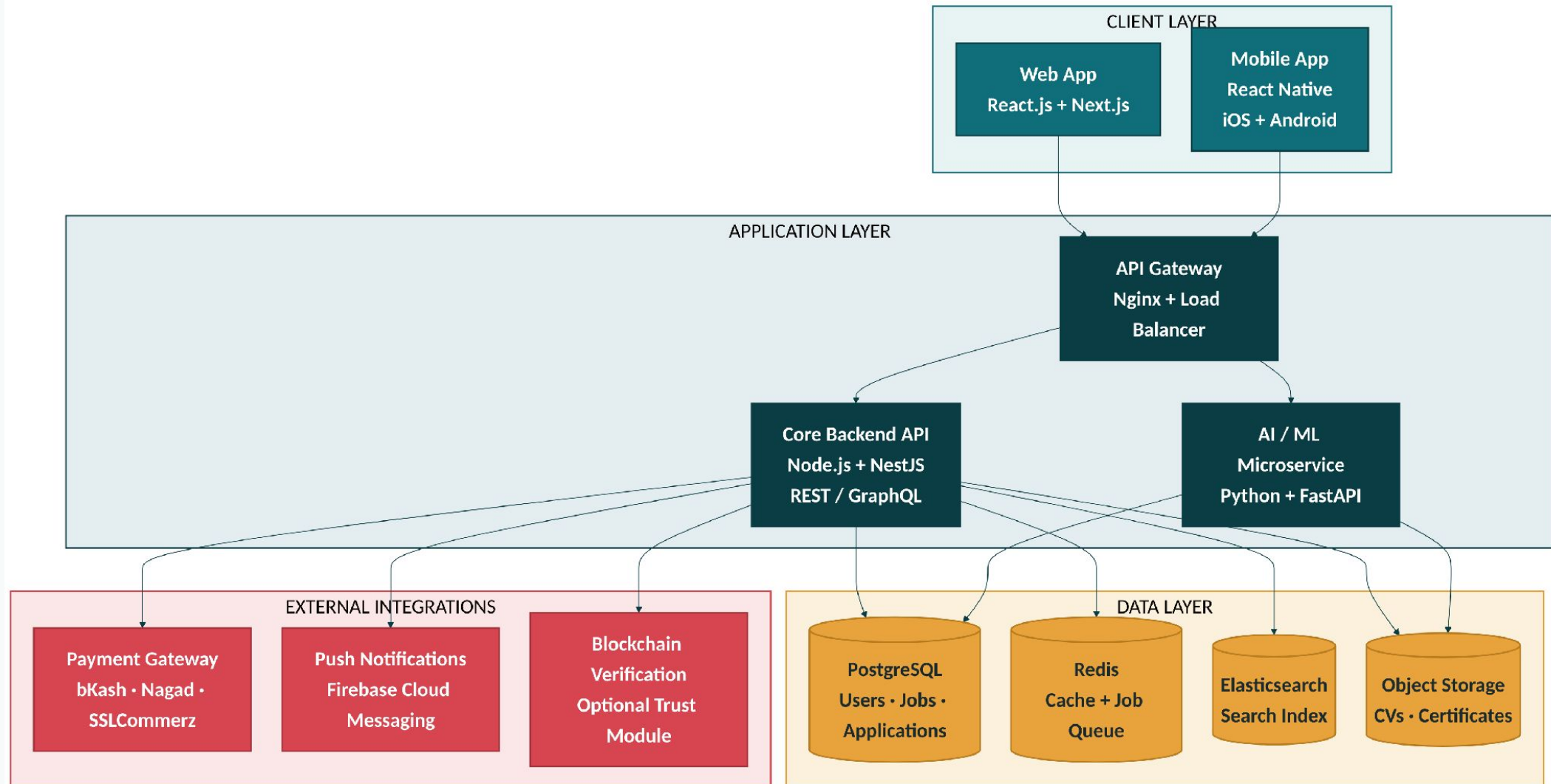
CI/CD to staging then production



Iterate

Retrospective & backlog refine

System Architecture



Our AI/ML Strategy: Three Pillars of Trust & Growth



① AI CV Grooming

Parses each CV with NLP, compares it to target-role benchmarks using sentence embeddings, and returns a Match Score with concrete improvement tips.

spaCy NER · Sentence-BERT · cosine similarity



② Smart Job Matching

Represents alumni and job posts as vectors and ranks the best-fit opportunities for each alumnus — and the best-fit candidates for each company.

TF-IDF / embeddings · content-based filtering



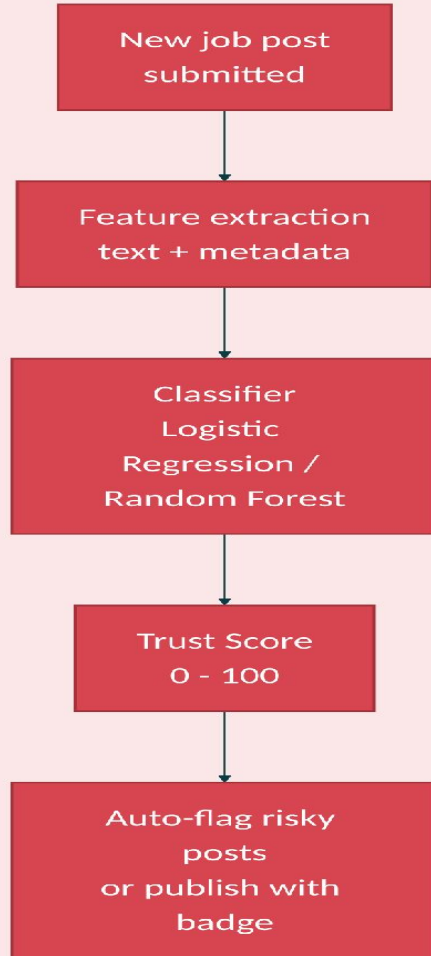
③ Job Trust & Fraud Detection

Classifies every new job post as legitimate or high-risk before it goes live, and assigns a visible Trust Score to protect alumni from scams.

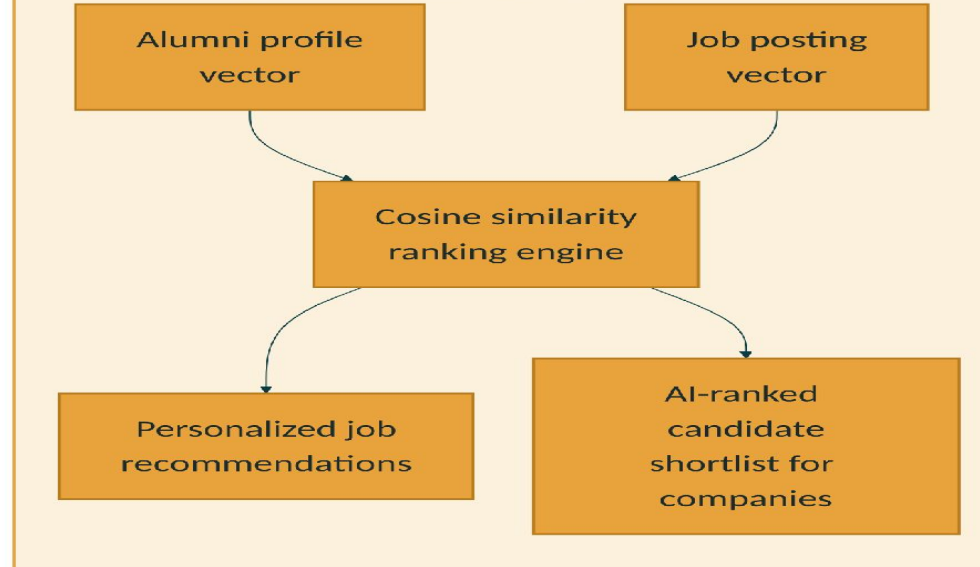
Logistic Regression / Random Forest → BERT roadmap

AI / ML Pipeline

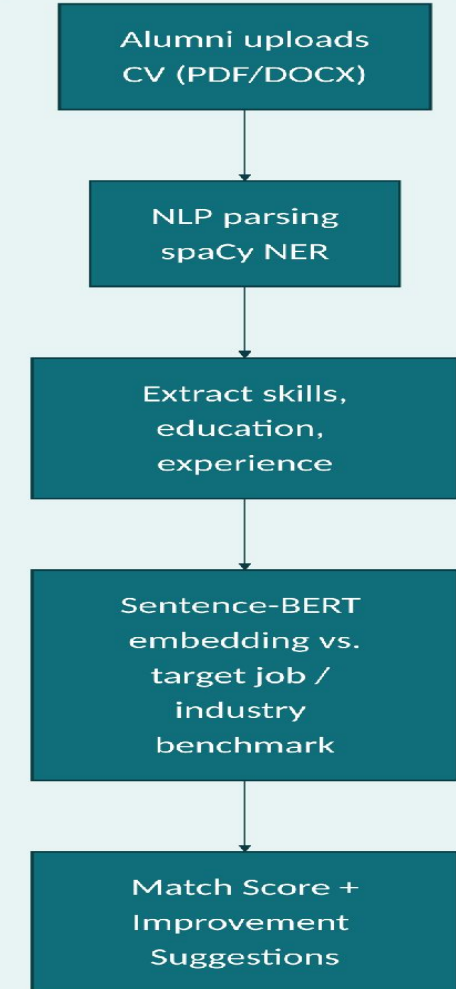
③ JOB TRUST & SCAM DETECTION



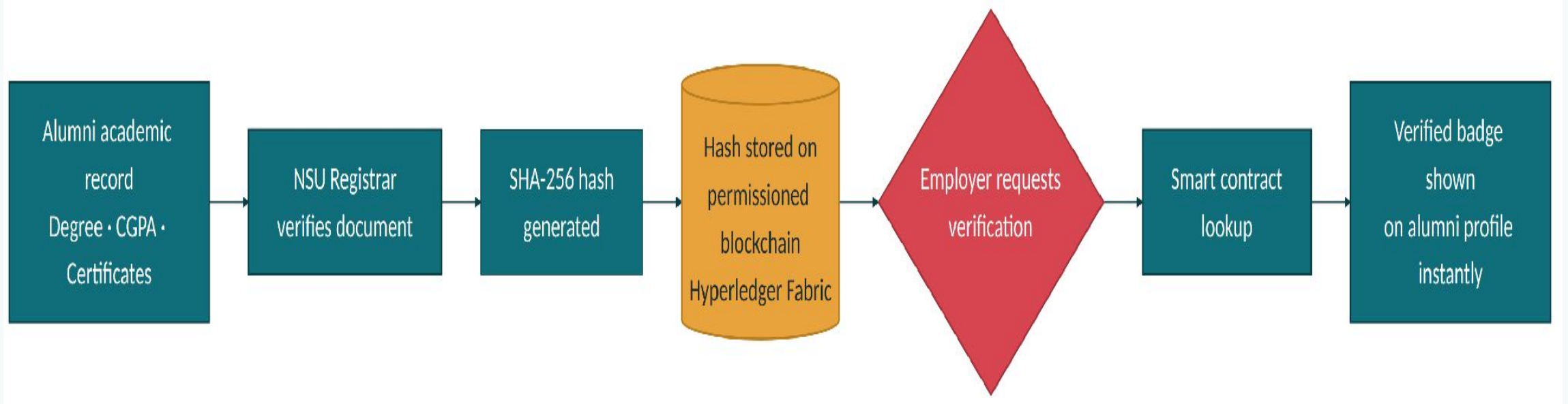
② SMART JOB-ALUMNI MATCHING



① AI CV GROOMING ASSISTANT



Blockchain Credential Verification



Tamper-proof

Degree and CGPA hashes on a permissioned ledger cannot be silently altered.



Instant lookup

Employers verify a credential in seconds instead of emailing the registrar.



Stretch goal

Built if the timeline allows — the core platform is fully functional without it.

Where Our Data Comes From



Primary data

Alumni & Job-Market Survey

Collected via a Google Form circulated across NSU alumni groups, LinkedIn communities and partner companies. Captures profile, skills, job history and hiring preferences.

Target: 800–1,000+ responses



Kaggle (secondary)

Resume Dataset

A public collection of resumes spanning roughly 25 job categories, used to train the CV parsing and section-classification model behind CV Grooming.

~2,400+ labeled resumes



Kaggle (secondary)

Real / Fake Job Postings (EMSCAD)

The Employment Scam Aegean Dataset — the standard public benchmark for training the job-post fraud / trust classifier.

17,880 postings · 866 fraudulent · 18 features

Dataset Features

Alumni Survey

- Name, Graduation Year, Degree
- CGPA, Credits Completed
- Skills & Certifications
- Current Job Title, Industry
- Years of Experience
- Desired Role / Career Goal

Resume Dataset

- Category (~25 job fields)
- Raw Resume Text
- Extracted Skill Keywords
- Section Labels (Education, Experience, Projects...)
- Word / Section Length

Job Postings (EMSCAD)

- title, location, department
- salary_range, company_profile
- description, requirements, benefits
- telecommuting, has_company_logo,
- has_questions
- employment_type, required_experience,
- required_education
- industry, function, fraudulent (label)

Application Layer

Frontend (Web)



React.js, Next.js, Tailwind CSS

Mobile App



React Native — shared codebase for iOS & Android

Backend API



Node.js, NestJS / Express, REST + GraphQL

AI / ML Microservice



Python, FastAPI, scikit-learn, spaCy, Sentence-Transformers

Database & Storage



PostgreSQL, Redis (cache/queue), Elasticsearch, S3-compatible object storage

Blockchain (Optional)



Hyperledger Fabric or Solidity + Web3.js

DevOps, CI/CD & Quality Assurance

Containerization



Docker & Docker Compose for consistent dev/staging/prod environments

CI/CD Pipeline



GitHub Actions — lint, test, build and deploy automatically on every push

Hosting & Cloud



AWS / DigitalOcean with Nginx as reverse proxy and load balancer

Monitoring & Logging



Sentry for error tracking, Grafana + Prometheus for system dashboards

Automated Testing



Jest + React Testing Library, PyTest, Postman/Newman API tests, Cypress E2E

Version Control & PM



Git + GitHub branching strategy, Jira / Trello sprint boards

Our Design Approach

- Figma-based wireframing and interactive prototyping before any code is written.
- Usability testing with 5–8 real alumni and recruiters per round, feeding directly into the next sprint.
- A shared, mobile-first design system (colors, type, components) kept consistent across web and app.
- Accessibility checks — WCAG-AA color contrast and full keyboard navigation — built into every review.

Mockups will be attached as an appendix once the first prototyping round is complete.

DESIGN PROCESS



Research

Alumni & recruiter interviews



Wireframe

Low-fidelity flows in Figma



Prototype

High-fidelity, interactive screens



Usability Test

Moderated sessions, 5–8 users



Refine

Iterate before next sprint

Who CareerSetu Serves



NSU Alumni

Job seekers and career changers across all graduating batches, looking to grow or pivot their careers.



Final-Year Students

Soon-to-be alumni who build a groomed profile early and enter the job market ahead of graduation.



Companies & Recruiters

SMEs to large corporates seeking verified, credential-checked graduates without the noise of generic job boards.



University Career Services

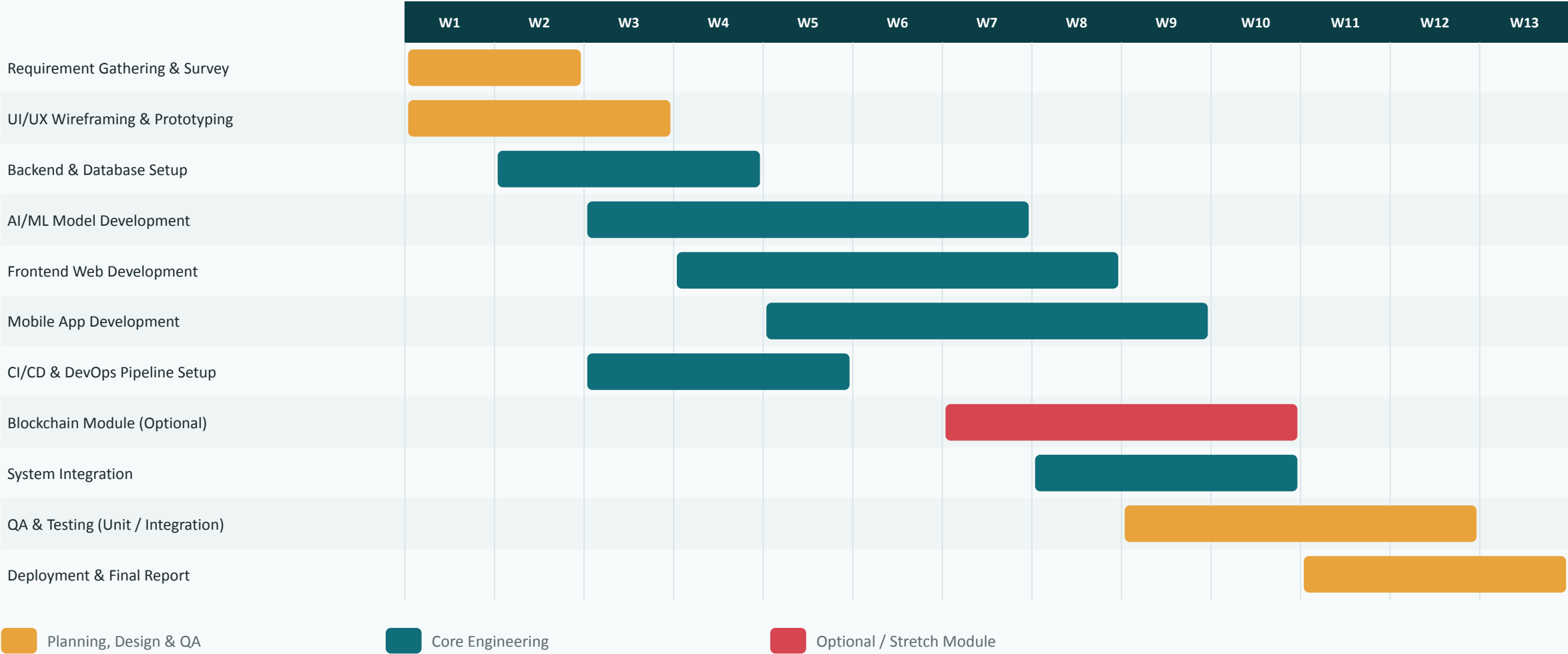
A data and engagement partner tracking alumni outcomes and strengthening industry ties.



A large, underserved segment

Roughly 29% of unemployed Bangladeshis aged 15–29 already hold a university degree (BBS Labour Force Survey, 2024). NSU alone graduates thousands of students every year — a ready-made, high-intent early user base for CareerSetu.

Gantt Chart — 13-Week Trimester Plan



Who Owns What

[we still haven't decided yet on who will be the lead of what but we will share the workload accordingly and work together to contribute and finish the project within the timeframe]



Member 1: To be added

UI/UX & Frontend (Web + Mobile) Lead

- Figma wireframes & prototypes
- React.js web app
- React Native mobile app
- Shared design system



Member 2: To be added

Backend & Database Lead

- Node.js / NestJS API
- PostgreSQL, Redis & Elasticsearch schema
- Payment gateway integration
- API security & auth



Member 3: To be added

AI / Machine Learning Lead

- CV parsing NLP model
- Job-alumni matching engine
- Fraud / trust classifier
- Dataset curation & training



Member 4: To be added

DevOps, QA & Blockchain Lead

- Docker & CI/CD pipeline
- Unit / integration / E2E testing
- Deployment & monitoring
- Optional blockchain module

All members collaborate across layers and support each other during integration, testing and crunch periods.

Sources & Datasets

- Vidros, S., Kolias, C., Kambourakis, G., & Akoglu, S. (2017). Automatic detection of online recruitment frauds: Characteristics, methods, and a public dataset. *Future Internet*, 9(1), 6. — origin of the EMSCAD dataset.
- Bansal, S. (2020). Real or Fake: Fake JobPosting Prediction [Dataset]. Kaggle.
- Kaggle. Resume Dataset — labeled resumes across ~25 job categories for CV parsing/classification research.
- Bangladesh Bureau of Statistics (BBS). (2024). Labour Force Survey 2024.
- The Business Standard. (2025, April). Educated but unemployed: Joblessness soars among graduates in Bangladesh.
- ScamCheck Bangladesh (scamcheck.gov.bd). Common Scams in Bangladesh — official cyber-scam awareness registry.
- The Business Standard. (2025). Youth unemployment in Bangladesh: A growing concern.



Thank You

CareerSetu — Questions & Discussion are welcome

CSE299 · North South University · Group 1